

ELEC70122: Machine Learning For Safety-Critical Decision Making

Lecture 5: Safe Reinforcement Learning and Constrained MDPs

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*Several slides have been used from Berenkamp and Krause, 2018.

So far we've focused on whether models are reliable



What treatment should a patient take?



How should a teacher assist a student?



How should a vehicle be driven?

The majority of safety-critical settings have *long-term consequences*.
Today, we move to systems that **act** in the world.

So far we've focused on whether models are reliable



How do we ensure safety in long-term decision-making?

What treatment should a patient take?

How should a teacher assist a student?

How should a vehicle be driven?

The majority of safety-critical settings have *long-term consequences*.
Today, we move to systems that **act** in the world.

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Outline for today's lecture

- Intro to Reinforcement Learning and Markov Decision Processes
- Specifying safety requirements and quantify risk
- Acting safely in *known* environments
- Acting safely in *unknown* environments
- Safe exploration (model-free and model-based)

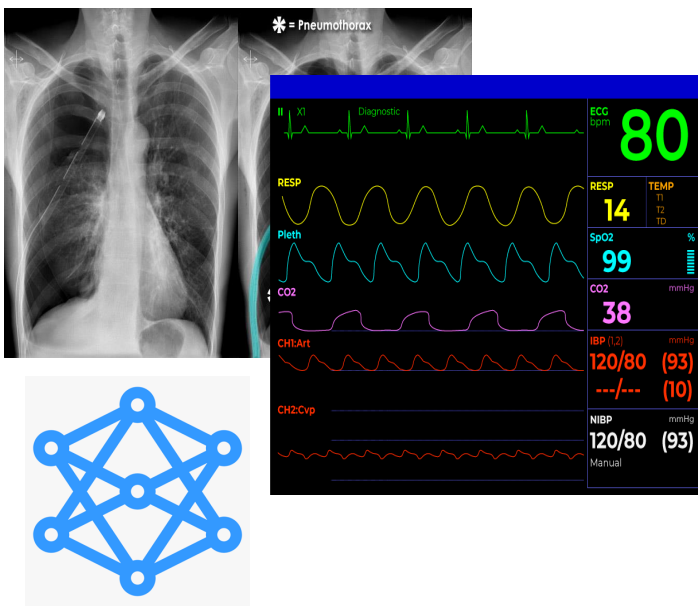
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Prediction != Decision



Suppose we have an ML model for sepsis risk prediction which predicts 0.82 risk of mortality.

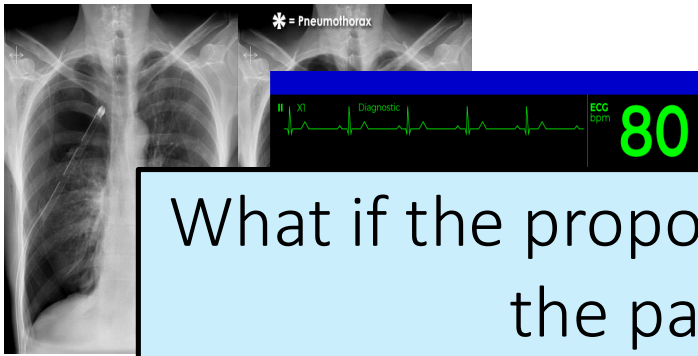
What should we do with this?

- Aggressive treatment?
- Wait?
- Escalate monitoring?

Prediction != Action

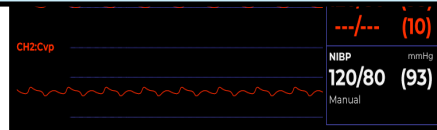
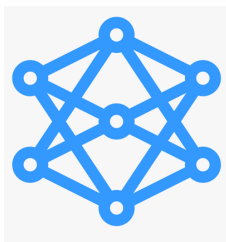
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Prediction != Decision



Suppose we have an ML model for sepsis risk prediction which predicts 0.82 risk of mortality.

What if the proposed action harms the patient?



- Wait?
- Escalate monitoring?

Prediction != Action

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Reinforcement Learning (RL)

Reinforcement Learning (RL)



Need to trade exploration & exploitation

Reinforcement Learning: An Introduction
R. Sutton, A.G. Barto, 1998



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Reinforcement Learning (RL)

- RL is modelled as a **Markov Decision Process (MDP)**:

$$(\mathcal{S}, \mathcal{A}, \mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}, \mathcal{T}, \gamma \in \mathbb{R})$$



Need to trade **exploration** & **exploitation**

- Goal is to maximize $\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{R}(s_t, a_t) \right]$

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Policies in RL and Policy Trajectories

- A **policy** is a sequence of actions $\pi(a|s)$ we execute from a particular state, or a decision rule mapping states to actions.
- Can be deterministic or **stochastic**
- Encodes decision strategy under uncertainty
- The optimal policy is learned to optimize long-term outcomes:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]$$

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Value Functions and Bellman Equations

We compute the expected long-term benefit of taking an action using the Bellman equations:

$$V^{\pi}(s) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]$$

$$Q^{\pi}(s, a) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]$$

- Value decomposes recursively
- Immediate reward + discounted future value
- Future consequences shape present decisions

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Reinforcement Learning Paradigms

Model-based RL

Transitions
 $p(S_t \mid S_{t-1}, A_{t-1})$

G-computation
MDP estimation

Value-based RL

Value/return
 $p(G_t \mid S_t, A_t)$

Q-learning
G-estimation

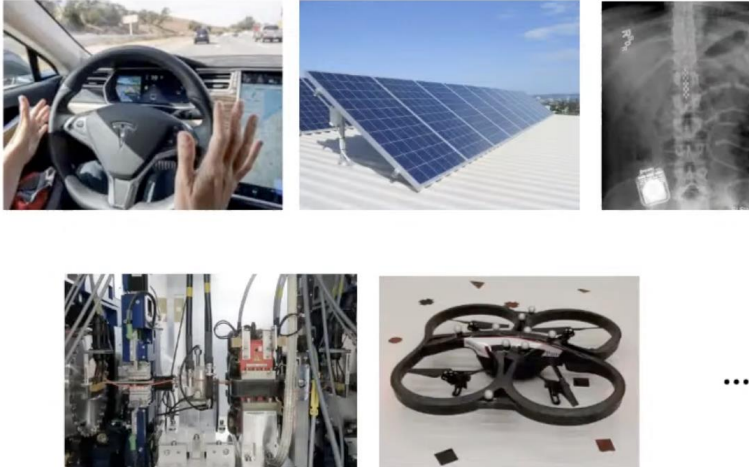
Policy-based RL

Policy
 $p(A_t \mid S_t)$

REINFORCE
Marginal structural models

How does safety break in RL?

Sequential decision-making introduces new safety risks.



- Unsafe exploration
- Error propagation
- Reward misspecification
- Catastrophic tail risk

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Unsafe exploration

Learning by interaction or trial-and-error can be harmful:

- Consequences of actions may be unknown
- Some actions may cause irreversible harm
- Trial-and-error may take you to unsafe states



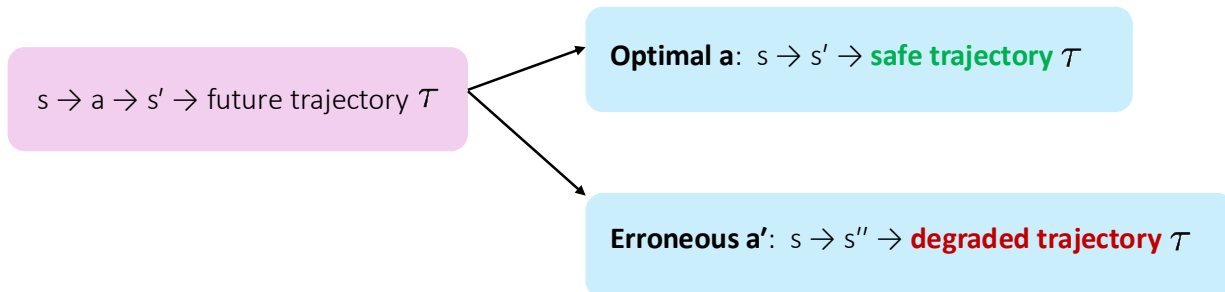
Exploration risk reflects epistemic uncertainty about action consequences.

Learning requires exploration, while safety requires caution¹⁴

Errors can propagate easily in sequential settings

- Since the Bellman Equations are recursive, any errors in the policy **cascade through time** through future states and consequences.

$$V^\pi(s) = \mathbb{E}_{a \sim \pi} \left[r(s, a) + \gamma \mathbb{E}_{s' \sim \mathcal{T}(\cdot|s, a)} [V^\pi(s')] \right]$$



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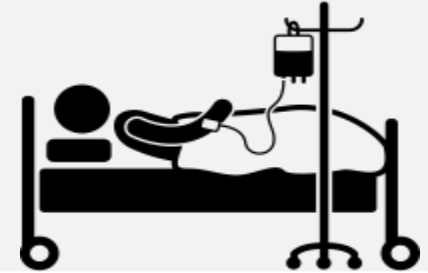
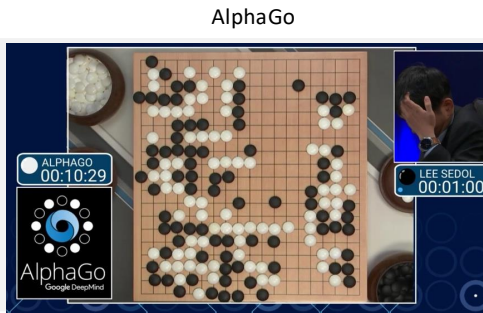
- Policy errors shift system state
- Future decisions operate on degraded states
- Harm compounds over time
- Failures may become irreversible

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Reward Misspecification

- Optimizing the **wrong objective** can produce unsafe behaviour

Problem Setting	AlphaGo	Healthcare
Interaction with Environment	Yes	No
Large, Relevant Training Samples	Yes	No
Simple objectives	Yes – Win/Lose	No – Healthy? Safe? Cost Effective?
Nonstationary	No	Yes

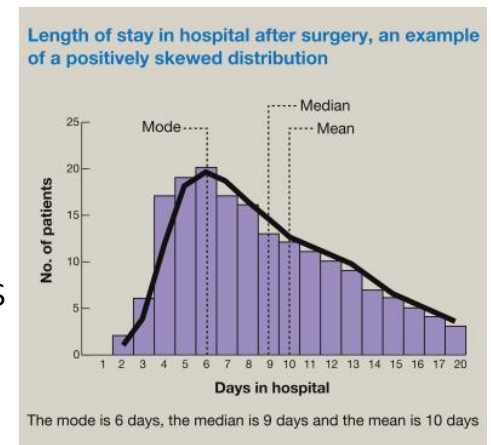


Catastrophic Tail Risk

- Standard RL optimizes for expected reward, not catastrophic risk:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{R}(s_t, a_t) \right]$$

- Expectation hides rare harms
- Rare failures are underweighted
- Safety is instead defined by worst-case outcomes
- Tail risk must be managed explicitly



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What does it mean to be safe?



Safety $\cong$ avoid bad trajectories (states/actions) $g(\{s_t, a_t\}_{t=0}^N) > 0$

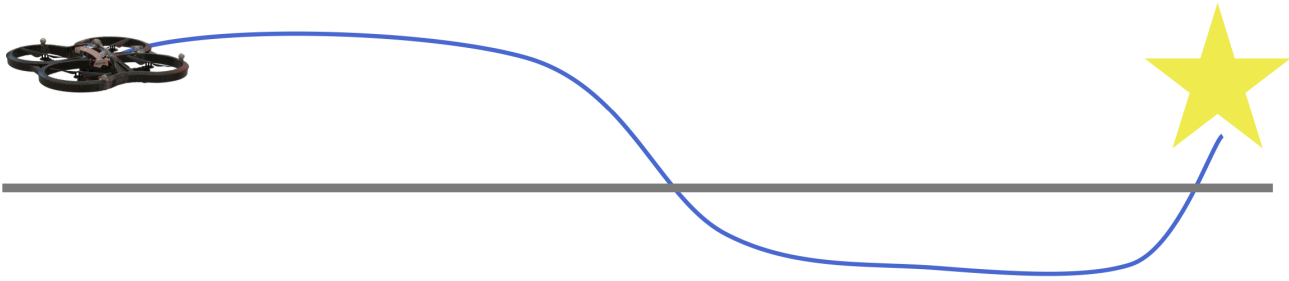
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Fix a policy $a_t = \pi(s_t, \theta)$

How do I quantify uncertainty and risk?

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Specifying safe behaviour



Is this trajectory safe?

$$g(\{s_t, a_t\}_{t=0}^N) = g(\tau) > 0$$

e.g. $g(\tau) = \min_{t=1:N} \Delta(s_t, a_t)$

Monitoring temporal properties of continuous signals

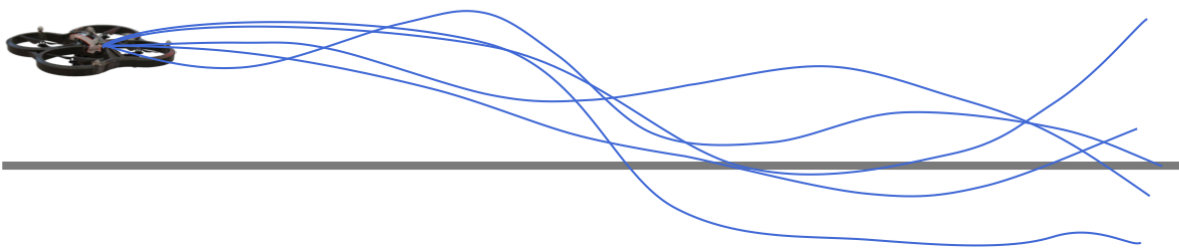
O. Maler, D. Nickovic, FT, 2004

Safe Control under Uncertainty

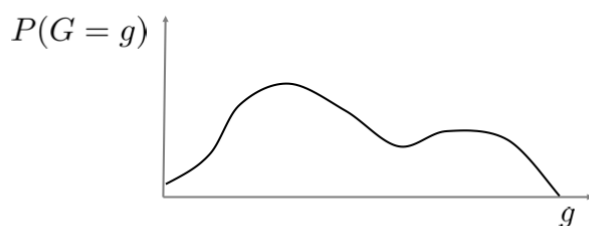
D. Sadigh, A. Kapoor, RSS, 2016

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Stochastic environment / policy



Safety function $g(\tau) \geq 0$ is now a random variable G



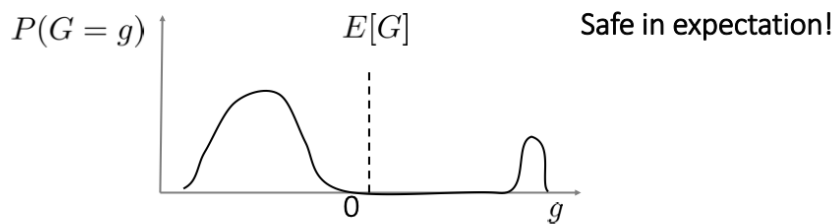
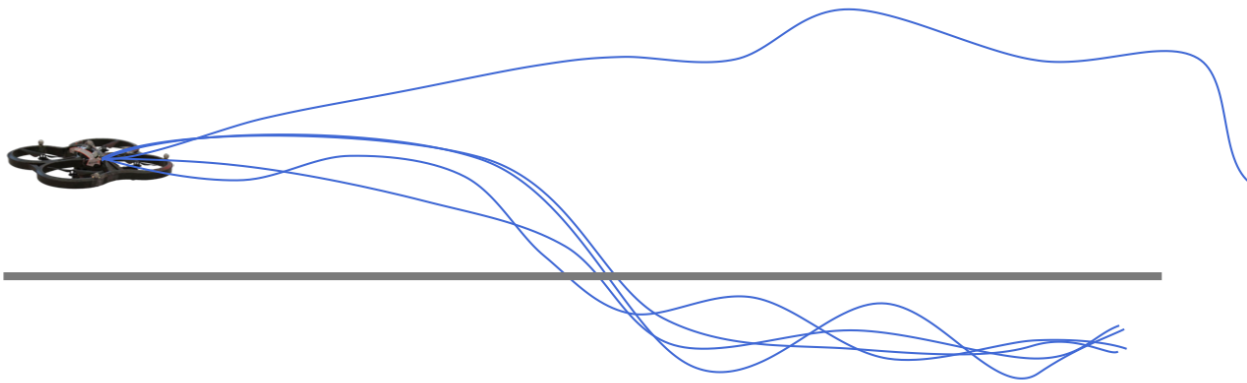
Expected safety

$$E[G]$$

;

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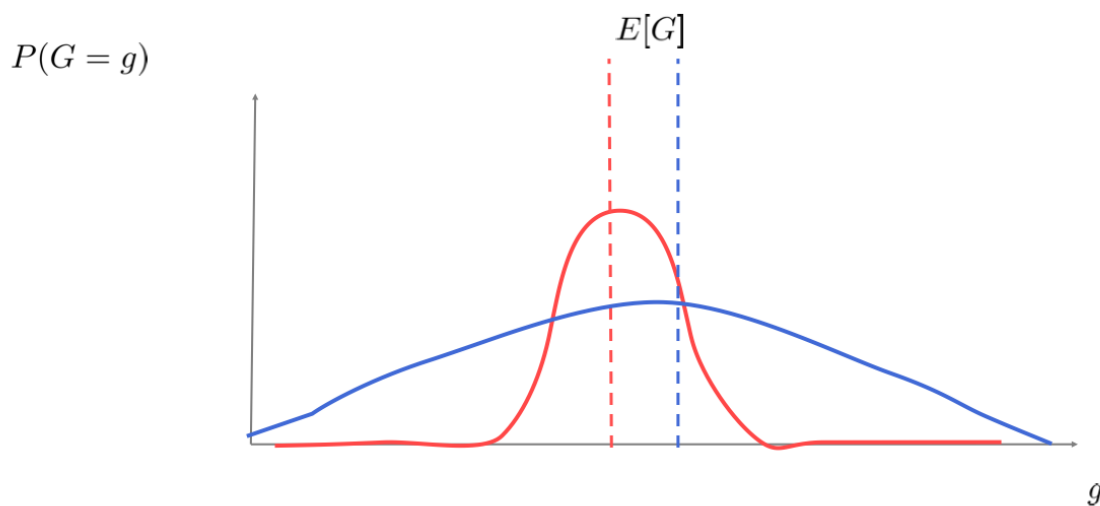
Expected safety can be misleading



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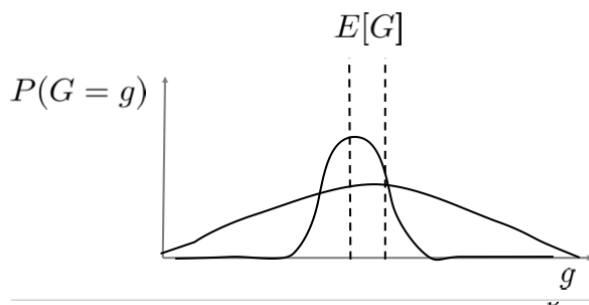
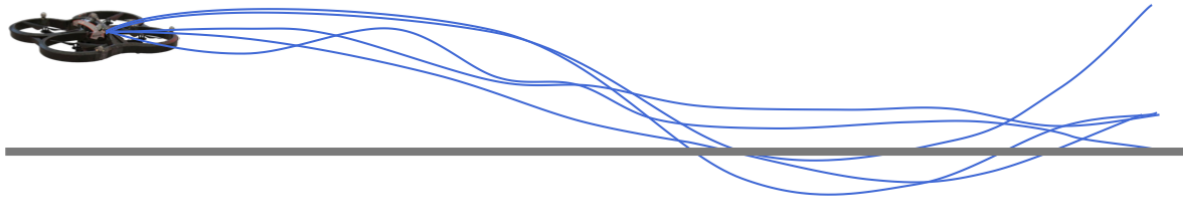
Expected safety can be misleading

Expected safety and variance



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Risk sensitivity

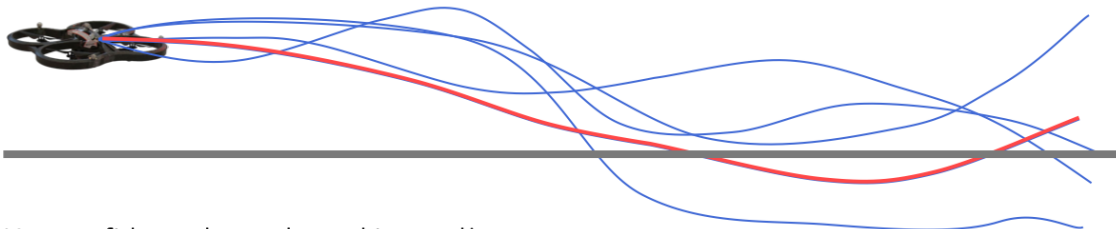


$$\mathbb{E}[e^{\tau G}] \propto \mathbb{E}[G] + \tau \mathbb{E}[G^2] + \mathcal{O}(\tau^2)$$

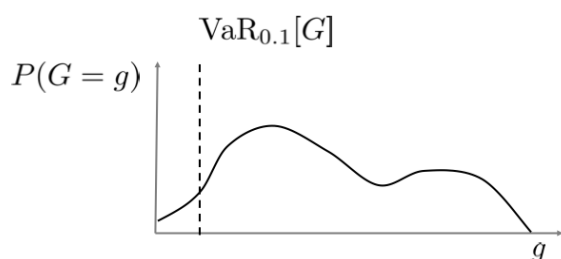
Even at low variance, a significant amount of trajectories may still be unsafe.

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Value-at-risk



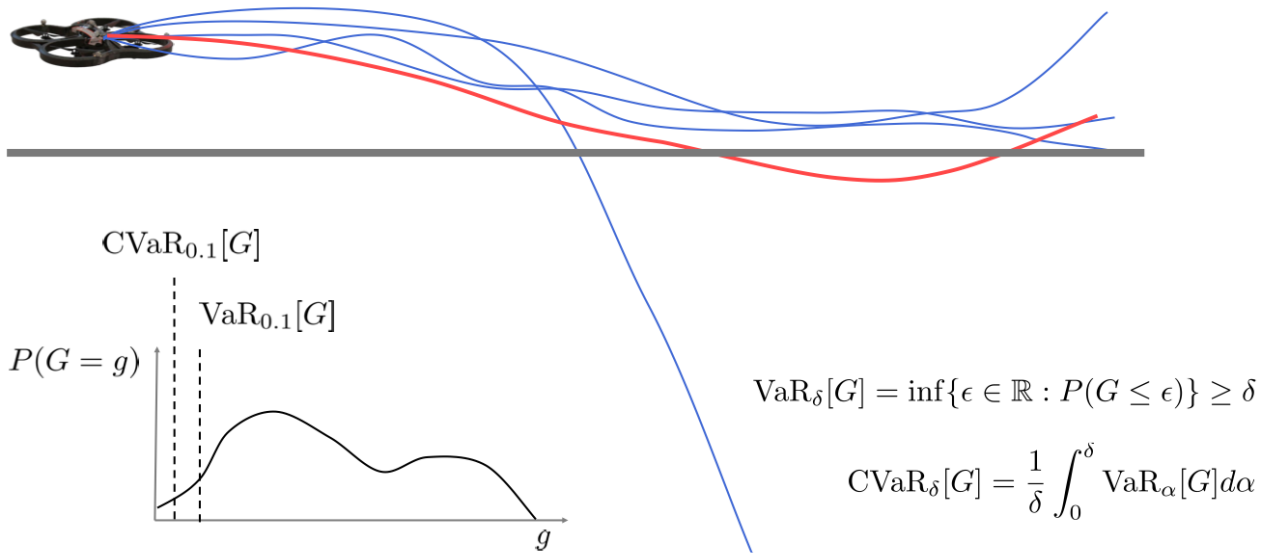
Use confidence lower-bound instead!



$$\text{VaR}_{\delta}[G] = \inf\{\epsilon \in \mathbb{R} : P(G \leq \epsilon)\} \geq \delta$$

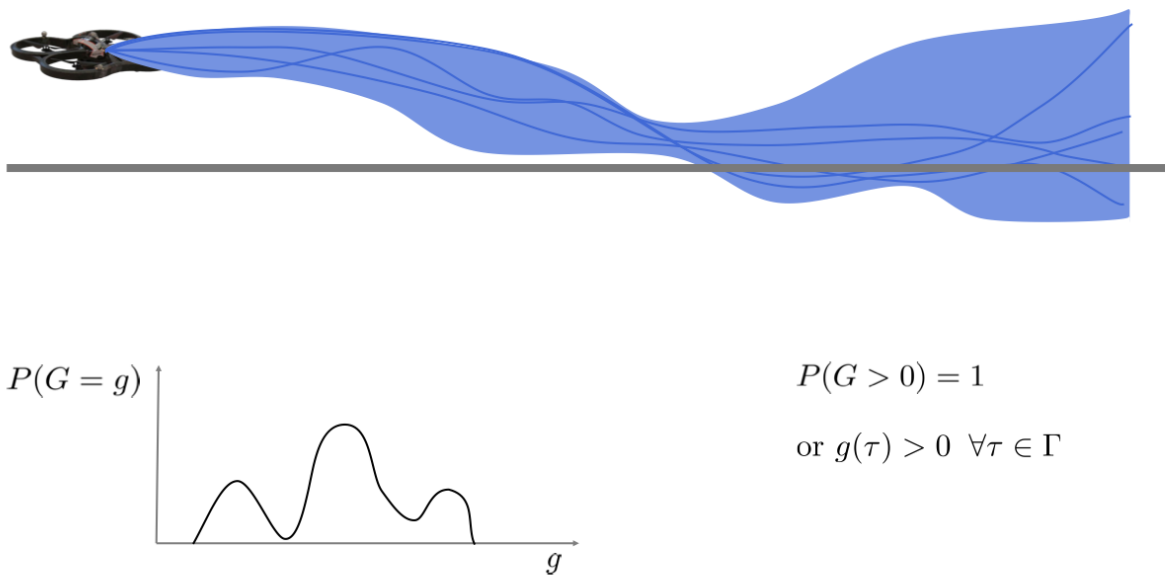
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Conditional value-at-risk



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Worst-case



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Notions of safety

Stochastic

Expected risk $E[G]$

Moment penalized $E[e^{\tau G}]$

Value at risk $\text{VaR}_{\delta}[G] = \inf\{\epsilon \in \mathbb{R} : P(G \leq \epsilon)\} \geq \delta$

Conditional value at risk $\text{CVaR}_{\delta}[G] = \frac{1}{\delta} \int_0^{\delta} \text{VaR}_{\alpha}[G] d\alpha$

Worst--case

$g(\tau) > 0 \quad \forall \tau \in \Gamma$

→ Robust Control

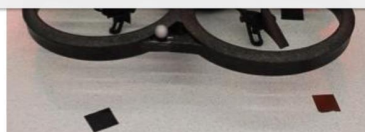
→ Formal verification

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Towards Safe RL

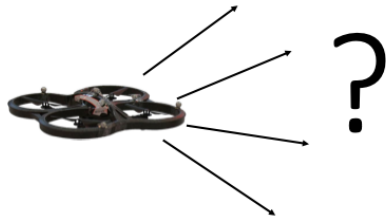


How can we *learn* to act
safely in *known* environments?



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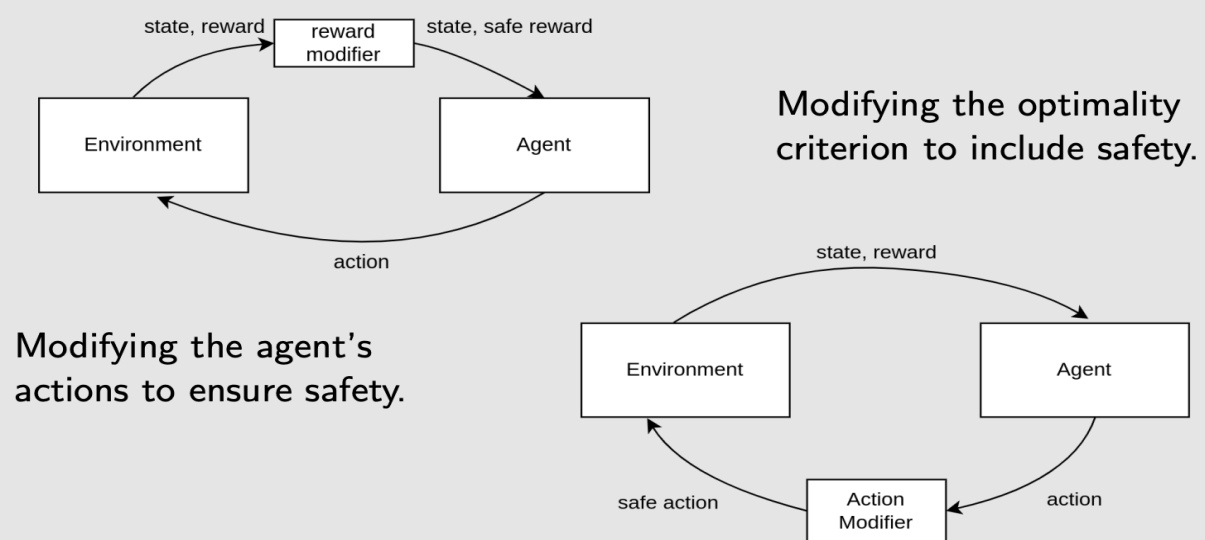
How to start acting safely?



No knowledge!
Now what?

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Either constrain the actions or modify optimality criterion



⁶Javier Garcia and Fernando Fernandez. "A Comprehensive Survey on Safe Reinforcement Learning". In: *J. Mach. Learn. Res.* 16.1 (Jan. 2015). ISSN: 1532-4435.

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Acting in *known* model with safety constraints



Constrained Markov decision processes
Eitan Altman, CRC Press, 1999

Essentials of robust control
Kemin Zhou, John C. Doyle, PH, 1998

Robust control of Markov decision processes with uncertain transition matrices
Arnab Nilim, Laurent El Ghaoui, OR, 2005

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Acting in *known* model with safety constraints



Key challenge: Don't know the consequences of actions!

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Constraining Actions to Safe Actions in Domain

$$a \in \mathcal{A}_{safe}(s)$$

Only actions in the safe set are allowed in state s .

- Restrict which actions can be taken
- Remove unsafe actions from consideration
- Enforced at decision time.

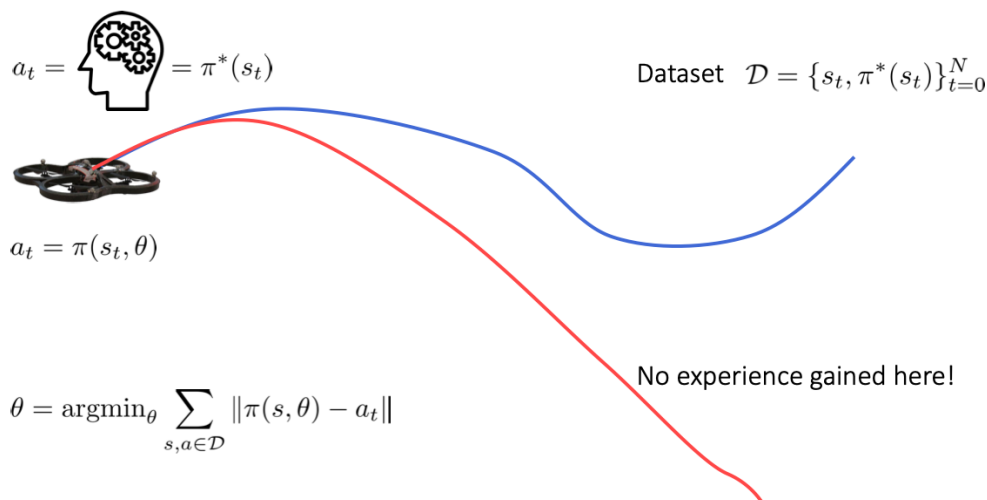
Examples

- Dosage limits in treatment
- Torque limits in robotics
- Speed bounds in autonomous driving
- Voltage constraints in power grids

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Constraining Actions with Imitation Learning

- Imitation learning can act as a safety constraint.



- Restricts actions to **demonstrated behavior**
- Avoids unsafe trial-and-error exploration
- Provides safe initial policy

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Constraining Actions with Imitation Learning

- Imitation learning can act as a safety constraint.

Note that Imitation learning allows for safety only within demonstration distribution. Unseen states may *still* lead to unsafe actions.

$a_t = 4$



$a_t = \pi$

$$\theta = \operatorname{argmin}_{\theta} \sum_{s, a \in \mathcal{D}} \|\pi(s, \theta) - a_t\|$$

No experience gained here!

- Provides safe initial policy

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Imitation learning algorithms

Data aggregation

Generate state sequence with policy $\pi(s_t, \theta)$

$$D \leftarrow D \cup \{(s_t, \pi^*(s_t))\}_{t=1}^T$$

$$\theta = \operatorname{argmin}_{\theta} \sum_{s, a \in \mathcal{D}} \|\pi(s, \theta) - a_t\|$$

Search-based Structured Prediction

Hal Daume III, John Langford, Daniel Marcu, ML, 2009

Efficient Reductions for Imitation Learning

Stephan Ross, Drew Bagnell, AISTATS 2010

Policy aggregation

$$\pi_0 = \pi^*$$

Generate state sequence $\mathcal{D}$ with policy π_i

$$\theta_{i+1} = \operatorname{argmin}_{\theta} \sum_{s, a \in \mathcal{D}} \|\pi(s, \theta) - a_t\|$$

$$\pi_{i+1}(s) = \alpha_0 \pi^*(s) + \sum_{j=1}^{i+1} \alpha_j \pi(s, \theta_j)$$

A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

Stephane Ross, Geoffrey J. Gordon, J. Andrew Bagnell, 2011

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Safe Imitation Learning

Only apply action from learned policy when

$$\|\pi(s_t, \theta) - \pi^*(s_t)\| \leq \epsilon$$

Query-Efficient Imitation Learning for End-to-End Autonomous Driving
Jiakai Zhang, Kyunghyun Cho, AAAI, 2017

$$\text{Var}[\pi(s_t, \theta)] \leq \gamma$$

EnsembleDagger: A Bayesian Approach to Safe Imitation Learning
Kunal Menda, Katherine Driggs-Campbell, Mykel J. Kochenderfer, arXiv2018



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What to do with this initial policy?

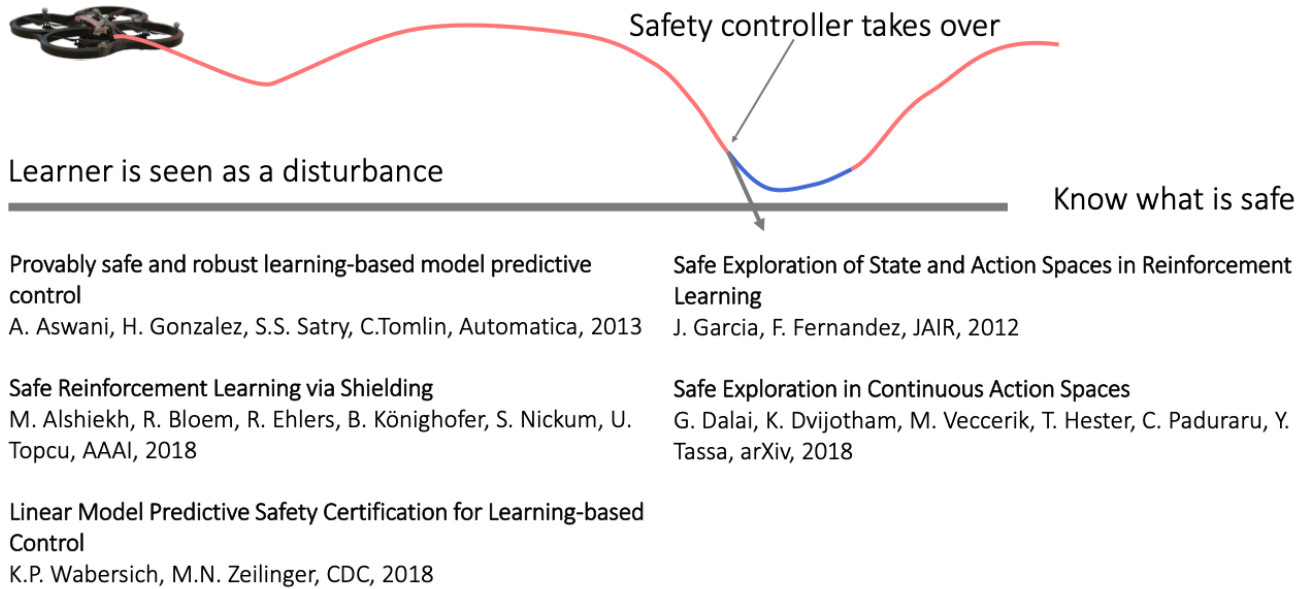
Can find an initial, safe policy based on domain knowledge.

How to improve?



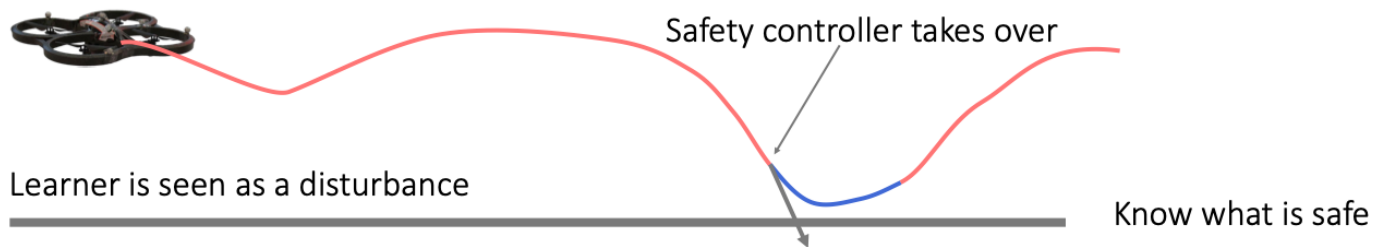
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Prior knowledge as a backup for learning



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Prior knowledge as a backup for learning



Need to know what is unsafe in advance.

Without learning, need significant prior knowledge.

The learner does not know what's happening!

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Safety as improvement in performance (Expected safety)

Initial, stochastic policy $\pi(s, \theta_b)$

Performance
$$J(\theta) = \mathbb{E}_{s_t \sim \rho(\theta)} \left[\sum_{t=1}^T \gamma^t r_t(s_t) \right] = \mathbb{E}_{\tau \sim \rho(\theta)} [g(\tau)]$$

Safety constraint
$$\Pr(J(\theta) \geq J(\theta_b)) \geq 1 - \delta$$

Need to estimate $J(\theta)$ based only on data from $\pi(s, \theta_b)$

Next week's lecture will tell us how to do this

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Constrained MDPs

- Constrained MDPs use a modified MDP structure:

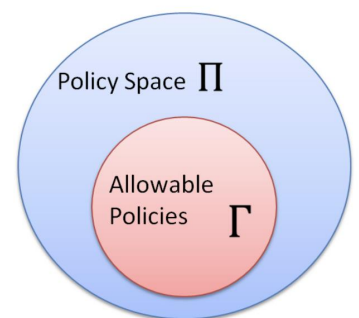
$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \mathcal{C}, \gamma)$$

- $\mathcal{C}$ specifies a **set of constraints applied to the policy**.

The general form of this criterion is:

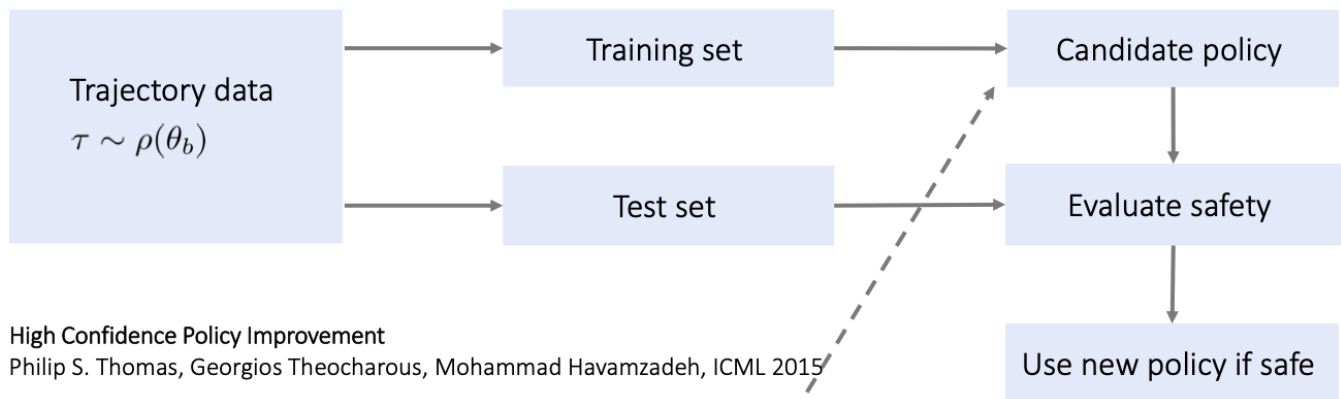
$$\max_{\pi \in \Pi} E_{\pi}(R) \text{ subject to } c_i \in C, c_i = \{h_i \leq \alpha_i\},$$

- c_i is the i -th constraint in C that the policy π must fulfill
- h_i is a function related with the return
- α_i is the threshold restricting the values of this function.



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Overview of expected safety pipeline



High Confidence Policy Improvement

Philip S. Thomas, Georgios Theodorou, Mohammad Havamzadeh, ICML 2015

Safe and efficient off-policy reinforcement learning

Remi Munos, Thomas Stepleton, Anna Harutyunyan, Marc G. Bellemare, NIPS, 2016

Constrained Policy Optimization

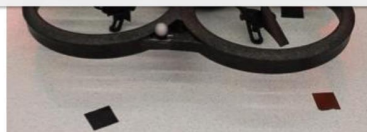
Joshua Achiam, David Held, Aviv Tamar, Pieter Abbeel, ICML, 2017

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Towards Safe RL



How can we *learn* to act
safely in unknown environments?



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Reinforcement learning (recap)

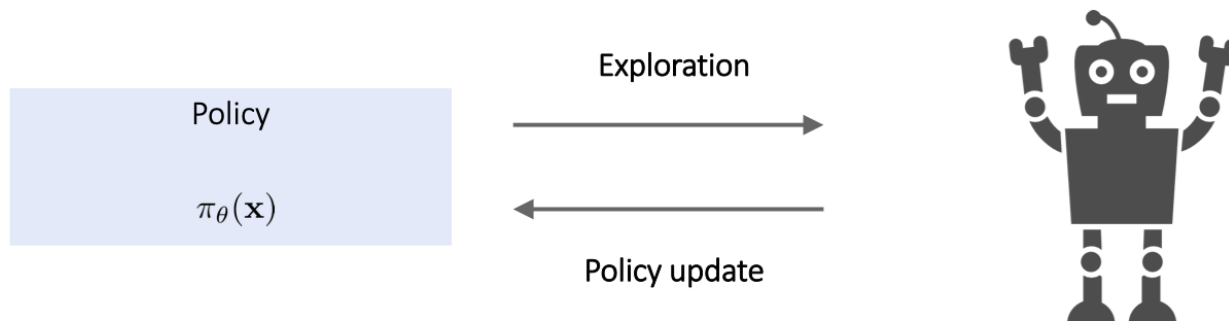


Image: Dorian, <https://iStock.com>

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Safe Reinforcement Learning

Statistical models to guarantee safety

Model-free

Model-based

$$a_t = \pi(s_t, \theta)$$

$$[s_{t+1}, r_t] \sim P(\cdot \mid s_t, a_t; \theta)$$

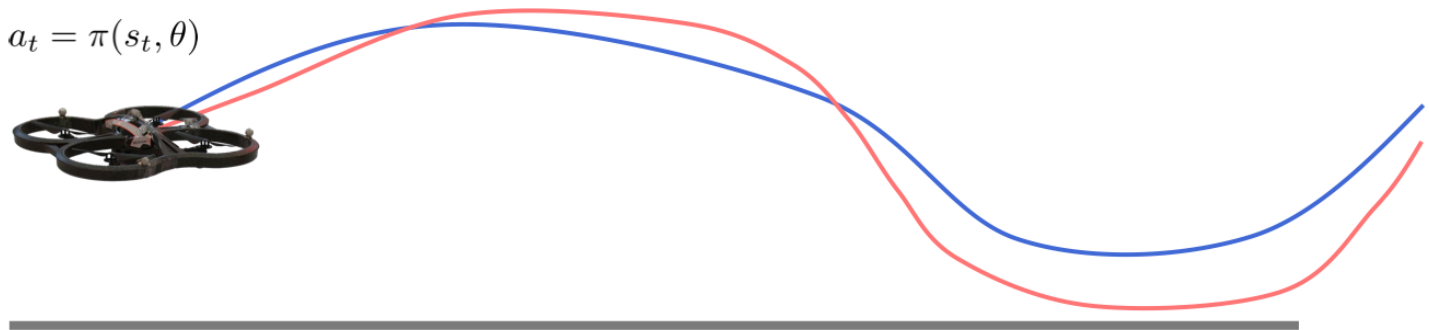
Estimate $J(\theta)$
and optimize

Estimate/identify,
then plan/control

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Model-free reinforcement learning

$$a_t = \pi(s_t, \theta)$$



Tracking performance

$$\max_{\theta} J(\theta)$$

Few, noisy experiments

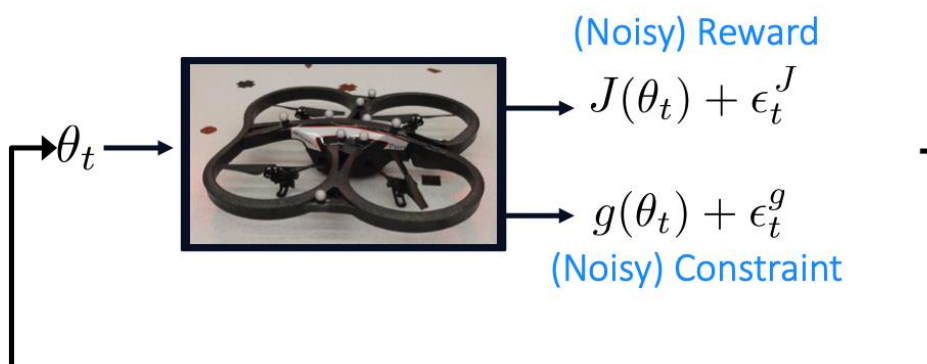
Safety constraint

$$g(\theta) \geq 0$$

Safety for all experiments

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Safe Policy Optimization

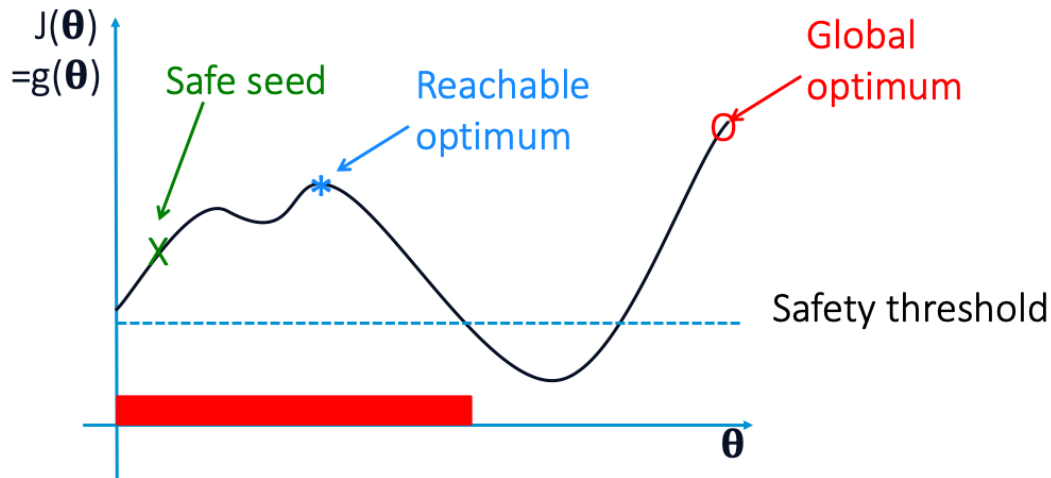


Goal: $\max_{\theta} J(\theta) \text{ s.t. } g(\theta) \geq 0$

Safety: $g(\theta_t) \geq 0 \text{ for all } t \text{ with probability } \geq 1-\delta$

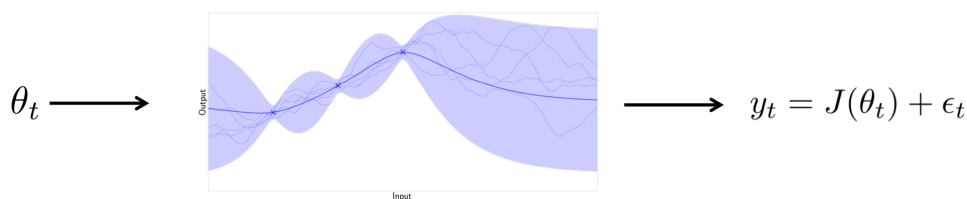
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Safe Policy Optimization Illustration

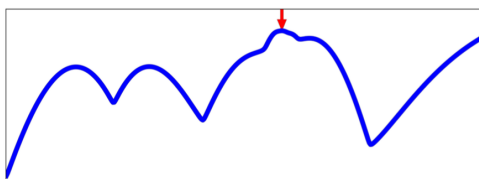


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Starting Point: Bayesian Optimization



Acquisition function



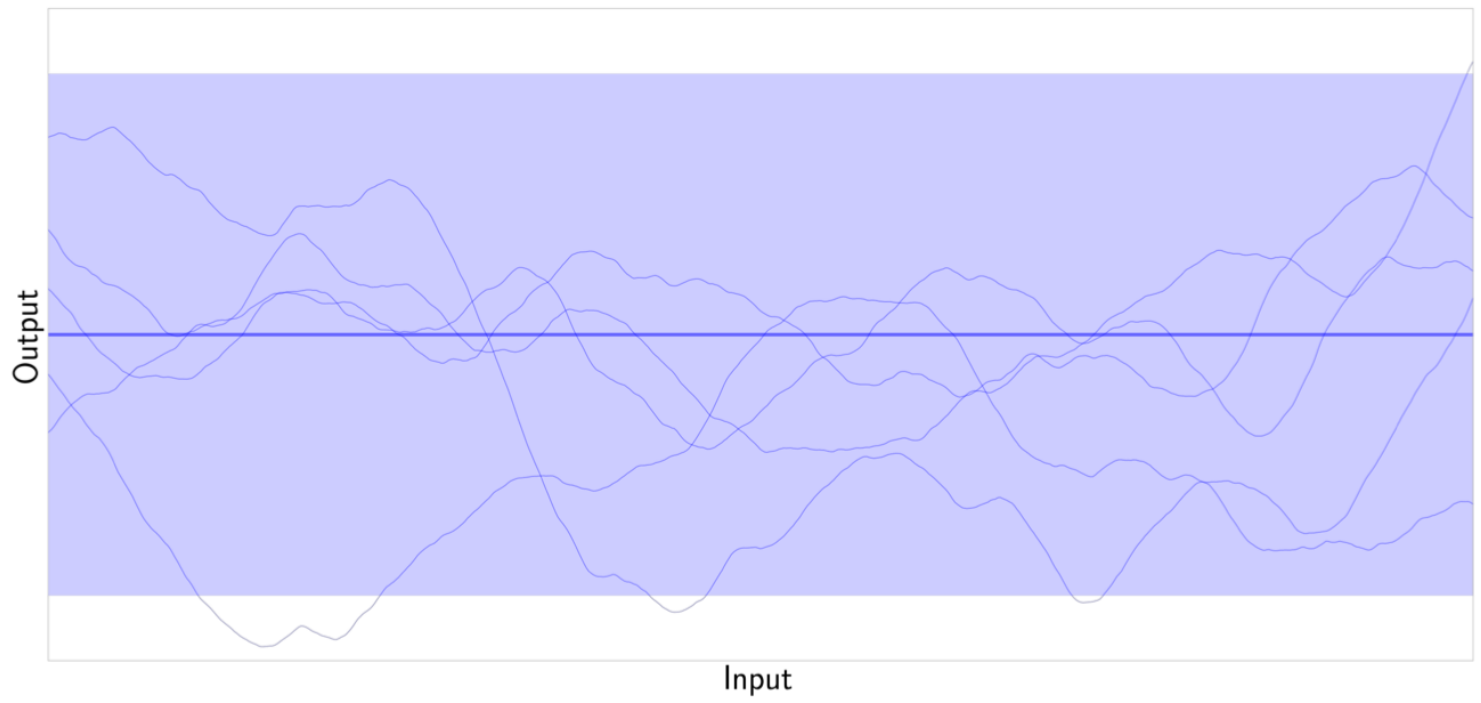
Expected/most prob. improvement [Moćkus *et al.* '78,'89]
 Information gain about maximum [Villemonteix *et al.* '09]
 Knowledge gradient [Powell *et al.* '10]
 Predictive Entropy Search [Hernández-Lobato *et al.* '14]
 TruVaR [Bogunovic *et al.* '17]
 Max Value Entropy Search [Wang *et al.* '17]

Constraints/Multiple Objectives

[Snoek *et al.* '13, Gelbart *et al.* '14, Gardner *et al.* '14, Zuluaga *et al.* '16]

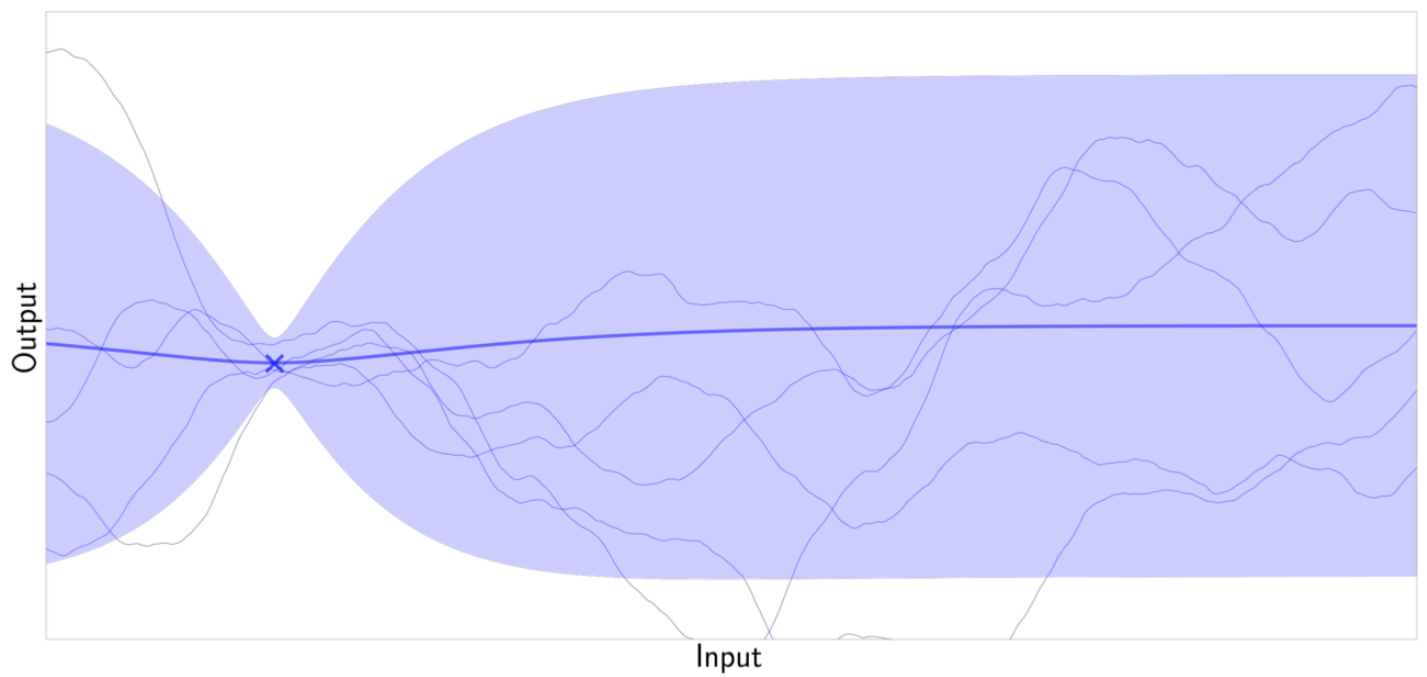
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Gaussian process



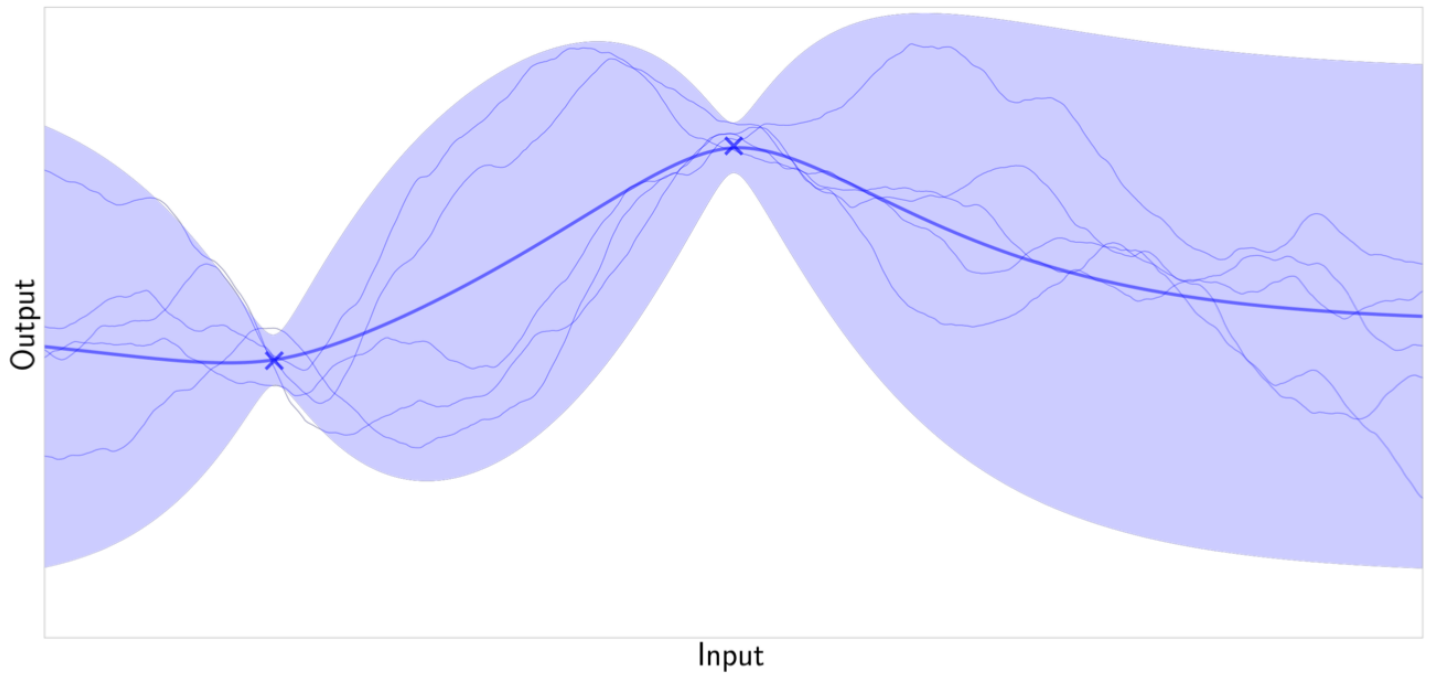
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Gaussian process



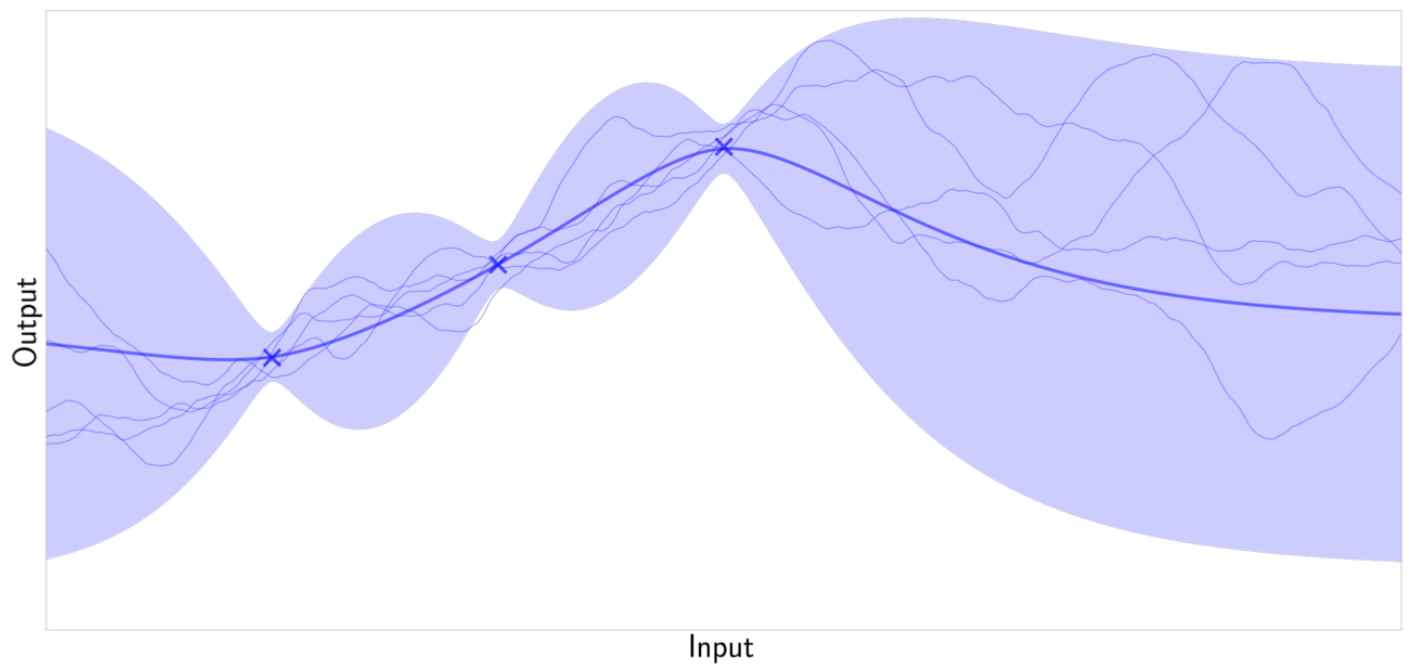
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Gaussian process



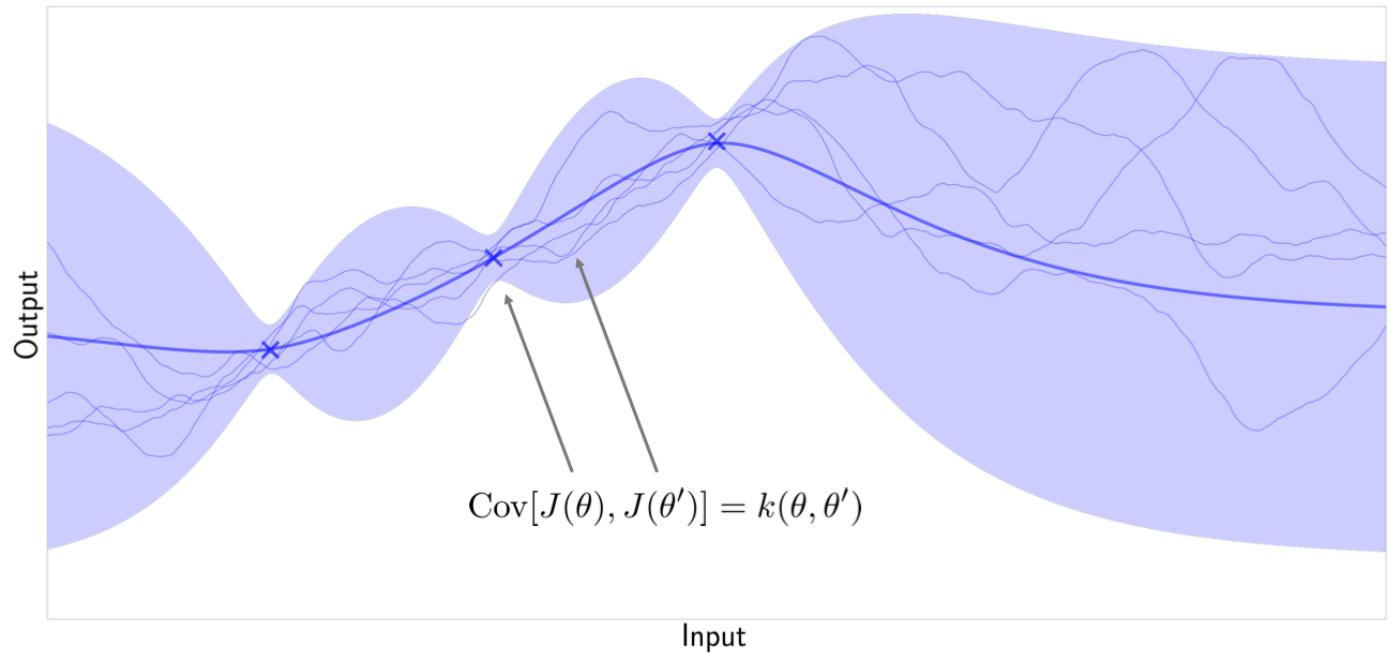
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Gaussian process



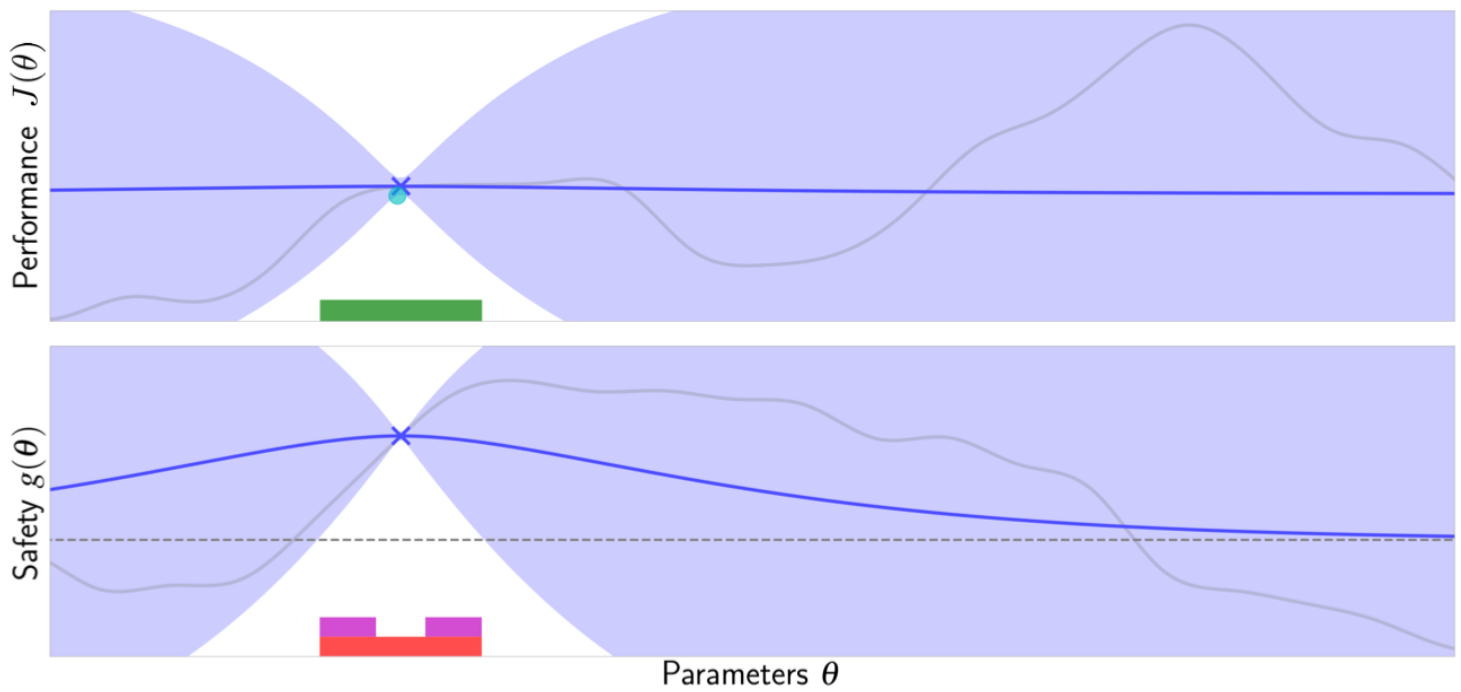
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Gaussian process



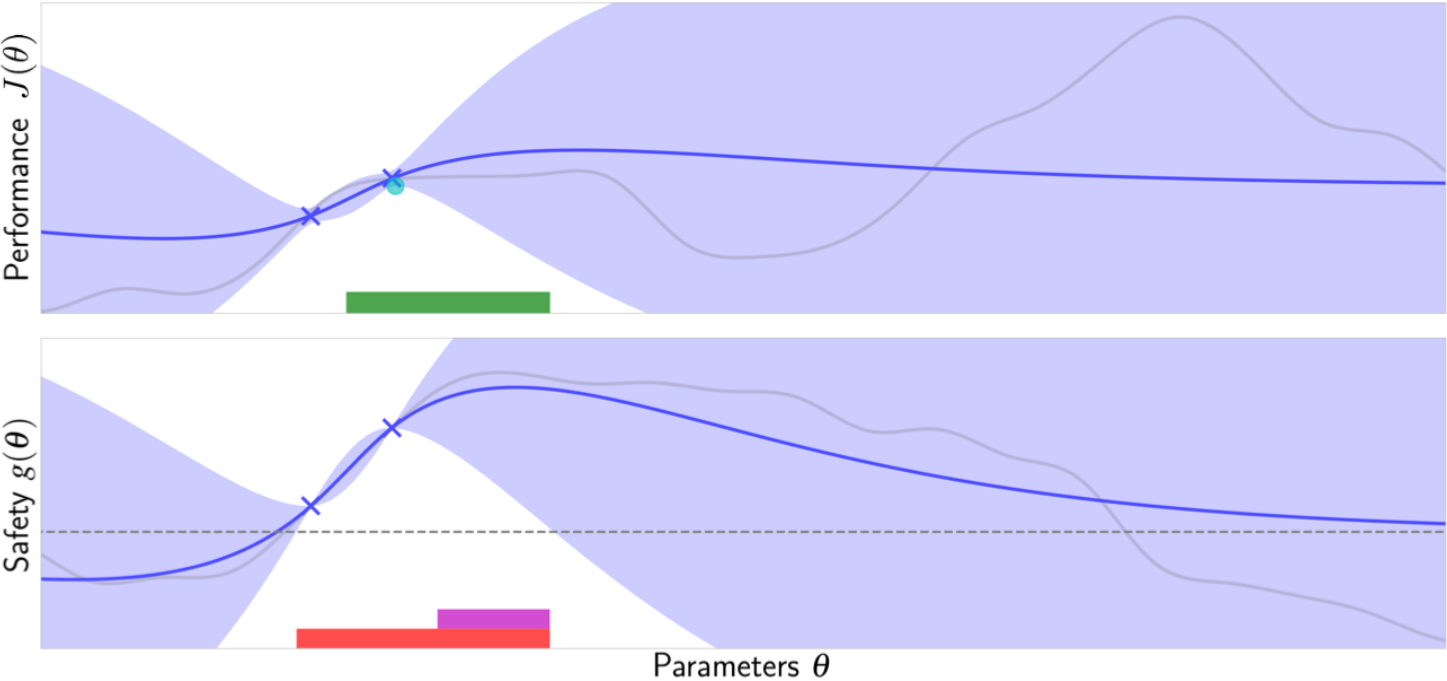
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SafeOPT: Constrained Bayesian optimization

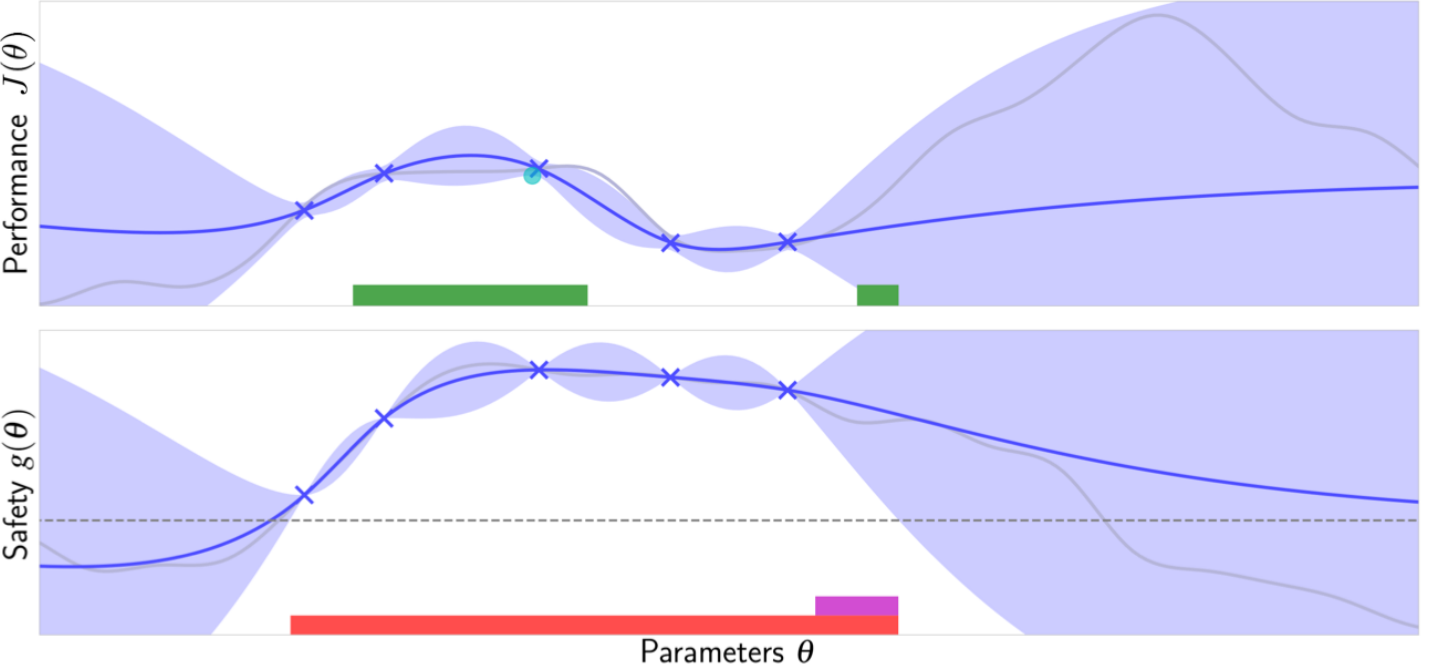


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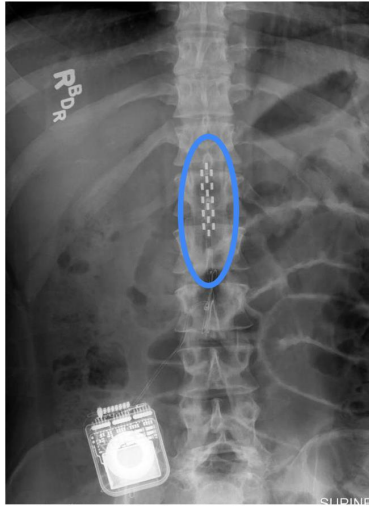
SafeOPT: Constrained Bayesian optimization



SafeOPT: Constrained Bayesian optimization



Therapeutic spinal cord stimulation



S. Harkema, The Lancet, Elsevier

Safe Exploration for Optimization with Gaussian Processes

Y. Sui, A. Gotovos, J. W. Burdick, A. Krause

Stagewise Safe Bayesian Optimization with Gaussian Processes

Y. Sui, V. Zhuang, J. W. Burdick, Y. Yue

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Safe Controller Tuning



Safe Controller Optimization for Quadrotors with Gaussian Processes

F. Berkenkamp, A. P. Schoellig, A. Krause, ICRA 2016

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Safe Bayesian Optimization (SafeOPT) guarantees

Theorem (informal):

Under suitable conditions on the kernel and on J, g , there exists a function $T(\epsilon, \delta)$ such that for any $\epsilon > 0$ and $\delta > 0$, it holds with probability at least $1 - \delta$ that

- 1) SAFEOPT **never makes an unsafe** decision
- 2) After at most $T(\epsilon, \delta)$ iterations, it found an **ϵ -optimal reachable** point

$$T(\epsilon, \delta) \in \mathcal{O} \left(\left(\|J\|_k + \|g\|_k \right) \frac{\log^3 1/\delta}{\epsilon^2} \right)$$

Safe Exploration for Optimization with Gaussian Processes
Y. Sui, A. Gotovos, J.W. Burdick, A. Krause

Safe Exploration for Active Learning with Gaussian Processes
J. Schreiter, D. Nguyen-Tuong, M. Eberts, B. Bischoff,
H. Markert, M. Toussaint

Bayesian Optimization with Safety Constraints: Safe and Automatic
Parameter Tuning in Robotics
F. Berkenkamp, A.P. Schoellig, A. Krause

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Safe Reinforcement Learning

Statistical models to guarantee safety

Model-free

Model-based

$$a_t = \pi(s_t, \theta)$$

$$[s_{t+1}, r_t] \sim P(\cdot | s_t, a_t; \theta)$$

Estimate $J(\theta)$
and optimize

Estimate/identify,
then plan/control

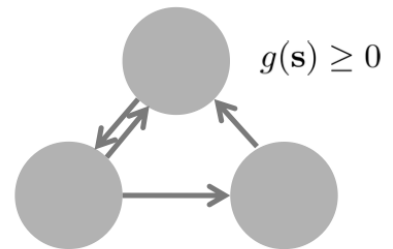
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From model-free bandits to MDPs

Bandit



Markov Decision Process

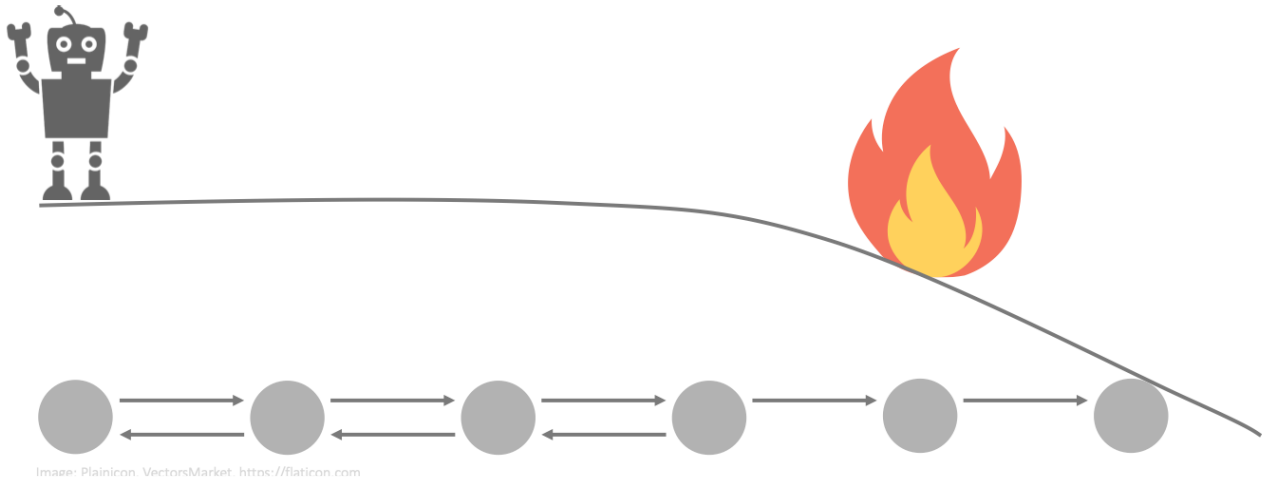


Can use the same Bayesian model to determine safety of states

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Challenges with long-term action dependencies

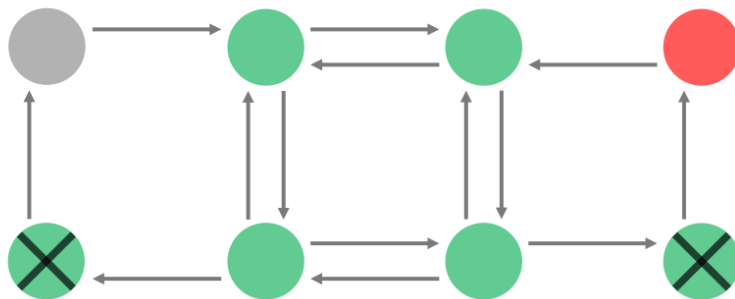
Non-ergodic MDP



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Rendering Exploration Safe

Rendering exploration safe



Safe Exploration in Markov Decision Processes
T.M. Moldovan, P. Abbeel, ICML, 2012

Safe Exploration in Finite Markov Decision Processes with Gaussian Processes
M. Turchetta, F. Berkenkamp, A. Krause, NIPS, 2016

Safe Exploration and Optimization of Constrained MDPs using Gaussian Processes
Akifumi Wachi, Yanan Sui, Yisong Yue, Masahiro Ono, AAAI, 2018

Exploration:

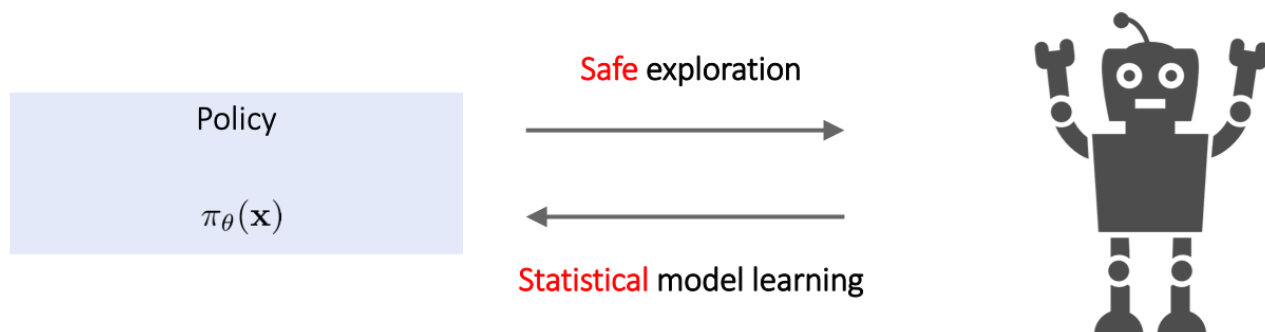
Reduce model uncertainty

Only visit states from which
the agent can recover safely

Safe Control under Uncertainty
D. Sadigh, A. Kapoor, RSS, 2016

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Safe Model-Based Reinforcement Learning



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A Bayesian Dynamics Model

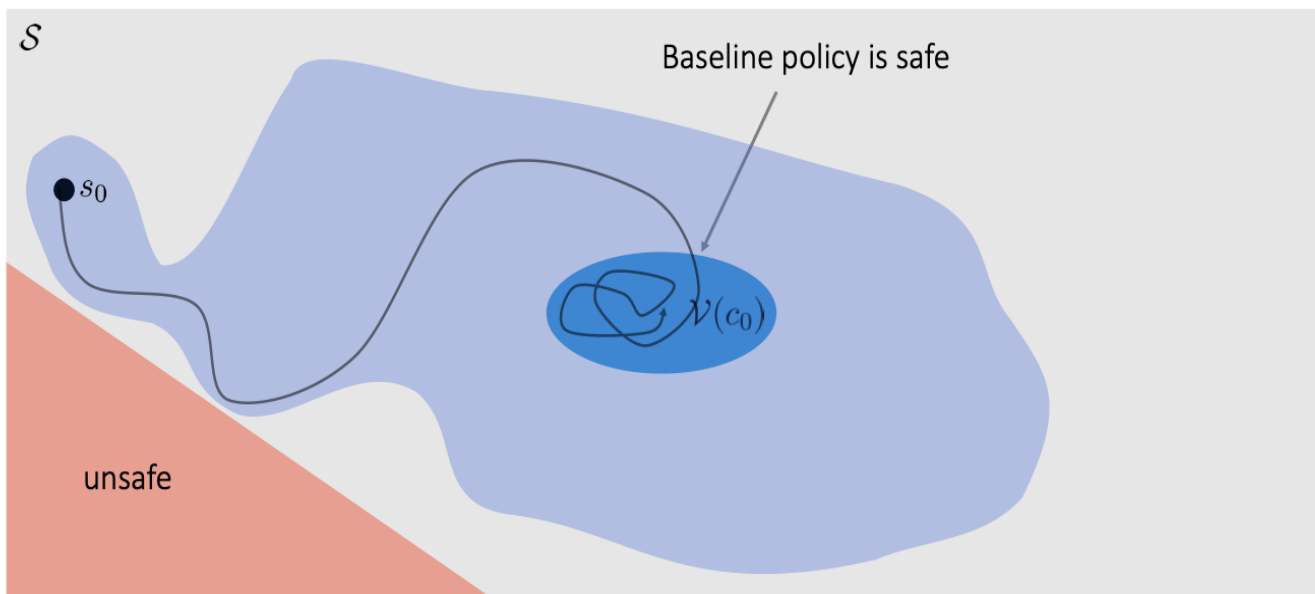
Dynamics

$$s_{t+1} = \underbrace{f(s_t, a_t)}_{a \text{ priori model}} + \underbrace{h(s_t, a_t)}_{\text{unknown model}}$$



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Region of Attraction



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